NASA Space Apps Challenge

FasalSathi

A NASA-powered crop rotation assistant for soil health and climate resilience

Team Mind Gamer 2.O
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Live demo: bho460.github.io/FasalSaathi

Why we built FasalSathi

My nana’s story

I grew up with my grandparents. My nana is a farmer who owns no land. He works other people’s fields, and the landowner takes 50% of the harvest, whether the crop made a profit or not.

I often saw crops grow poorly. Looking back, one reason may have been that the fields were never rotated. I did not know that then, and it still makes me sad.

NASA’s Open Science 101 course and newsletter led me to this challenge, and I remembered him.

Water

Over 55% of Uttarakhand’s cultivated area is rain-fed (ICAR).

Soil

Fertility is low to medium, with falling organic carbon (ICAR).

Risk

Farmers who rent land carry the risk of every poor season, so better choices matter.

From NASA data to rotation options

Farmer inputs

Zone, location, irrigation, slope, current rotation and priorities such as saving water or income.

1

NASA evidence

Temperature, rainfall, soil moisture, greenness and evapotranspiration for the location.

2

Compare 3 rotations

Existing plus two suggestions, with scores, ± uncertainty, risks and information gaps.

3

Scenario sliders: rainfall −40% to +30%, temperature up to +4 °C. Live app: bho460.github.io/FasalSaathi

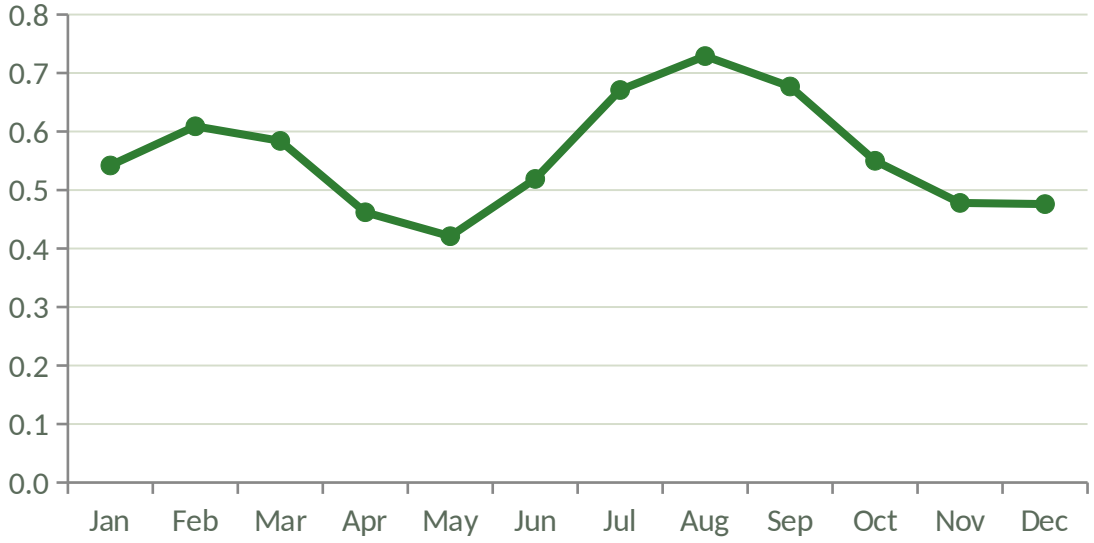
Four NASA datasets, harmonized to one point

NASA dataset	Resolution	Period	Used for
NASA POWER	~50 km	2004–2023, live API	Heat, frost, rainfall, water balance
MODIS NDVI (MOD13Q1)	250 m	2004–2023	Seasonal greenness, bare-field months
SMAP soil moisture	9 km	2015–2025	Dry months, drought-risk term
MODIS ET (MOD16A2)	500 m	2021–2023 only	Water-use evidence, not in scores

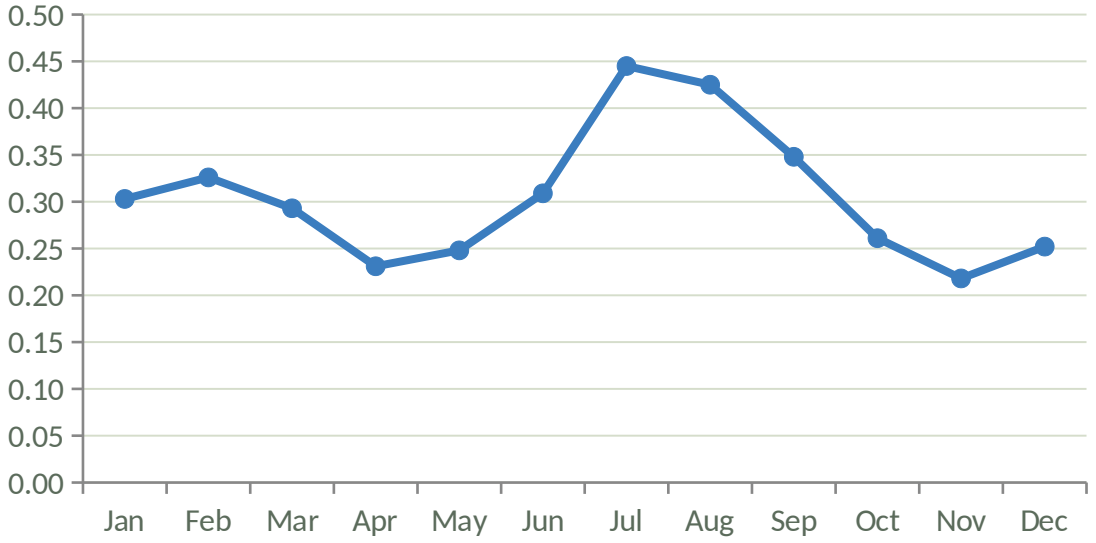
Different resolutions are aggregated to monthly values for one pilot field near Vikasnagar. Crop coefficients: FAO-56 (2025).

What NASA data shows on the pilot field

MODIS NDVI, monthly average 2004–2023



SMAP soil moisture (m^3/m^3), 2015–2025



Two crop seasons are visible from space: NDVI peaks in February (rabi) and August (kharif), with a dip in April–May after harvest. Soil is driest in November (0.22), the rabi sowing month.

Explainable and honest

Explainable scores

Five rule-based scores (water, soil, erosion, income, resilience) with ± ranges. No black box.

Data quality check

Our first point, in Dehradun city, peaked at NDVI 0.32. We recognised non-farmland and moved to a real farm pixel.

- Crop table values are expert-guided estimates; only Kc is from FAO-56
- One pilot field; ET covers only 3 years; the SMAP pixel is 9 km
- Always verify with the local KVK or agriculture officer

Impact and next steps

Who it helps

Farmers who carry the risk of every season can see what NASA data says about their location, compare rotation options, and ask better questions before choosing a crop.

Next steps

GRACE groundwater, SRTM slope, more fields and regions, a longer ET series, market prices, and validation with farmers and KVK experts.

We show patterns and possibilities. The final decision belongs to the farmer.